

Proof of chain rule

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1 Standard proof

In Bartle's Introduction to Real Analysis, we all have learnt that using Caratheodory's theorem, we can prove the chain rule rigorously. In fact, it is proved through the definition of derivatives. Here's the proof:

Theorem 1 (Caratheodory's theorem). *Let f be a continuous function defined on an interval I containing c . Then f is differentiable at c if and only if there exists a function φ on I that is continuous at c such that*

$$f(x) - f(c) = \varphi(x)(x - c)$$

Proof. ($\Rightarrow$) If f is differentiable at c , then $f'(c)$ exists. Define

$$\varphi(x) = \begin{cases} \frac{f(x) - f(c)}{x - c} & , x \neq c \\ f'(c) & , x = c \end{cases}$$

in which $\lim_{x \rightarrow c} \varphi(x) = f'(c) = \varphi(c)$. So φ is the continuous function required.

($\Leftarrow$) If φ is a function on I that is continuous at c such that

$$f(x) - f(c) = \varphi(x)(x - c),$$

then $\varphi(c)$ is well-defined and thus $f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \varphi(c)$ exists. $\square$

Throughout the proof of chain rule, we will apply Caratheodory's theorem, like in Bartle's.

Theorem 2 (Chain rule). *Let $f : I \rightarrow R$ and $g : D \rightarrow I$ be two differentiable functions, whose derivatives are denoted by f' and g' respectively. Let $h := f \circ g : D \rightarrow R$ be a composite function. Then h is differentiable on D and its derivative is defined as*

$$h' = (f' \circ g)g'$$

Proof. Fix any $c \in D$, $g(c)$ is well-defined. Then by Caratheodory's theorem, there exists continuous function φ on I such that

$$f(g(x)) - f(g(c)) = \varphi(x)(g(x) - g(c))$$

where $\varphi(c) = f'(g(c))$. Similarly, there exists another continuous function ϕ on D such that

$$g(x) - g(c) = \phi(x)(x - c)$$

where $\phi(c) = g'(c)$. Then

$$h'(c) = \lim_{x \rightarrow c} \frac{f(g(x)) - f(g(c))}{x - c} = \varphi(c)\phi(c) = f'(g(c))g'(c).$$

Since $c \in D$ is arbitrarily fixed, h' exists on D and $h' = (f' \circ g)g'$. $\square$

2 Proof using Taylor's expansion

Before the proof, we need the formulation of Taylor's expansion:

Theorem 3 (Rolle's Theorem). *Let f be a continuous function on an interval $[a, b]$ which is differentiable on (a, b) and $f(a) = f(b)$. Then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.*

Proof. $\forall x \in (a, b)$, define the secant function

$$m(\xi) := \frac{f(x) - f(\xi)}{x - \xi},$$

except when $x = \xi$, define $m(x) \rightarrow f'(\xi)$. Since $f'(\xi)$ exists, we can set

$$\varphi(x) = \begin{cases} m(x) & , x \neq \xi \\ f'(\xi) & , x = \xi \end{cases}.$$

Clearly, φ is continuous at ξ .

Suppose the claim is not true, we have different cases of m . Fix $\xi \in (a, b)$.

If $m \equiv 0$, then the case is trivial.

If m is strictly monotonic, then $f(b) = [f(b) - f(x)] + [f(x) - f(a)] + f(a) \neq f(a)$, which contradicts with our assumption.

If m has different signs on (a, ξ) and (ξ, b) , then by Intermediate Value theorem, $\varphi(\xi) = 0$. Hence, $f'(\xi) = 0$. $\square$

Theorem 4 (Cauchy Mean Value Theorem). *Let f and g be continuous functions on an interval $[a, b]$ which is differentiable on (a, b) . Then there exists $\xi \in (a, b)$ such that $\frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}$.*

Proof. Put $\varphi(x) := f(x) - \frac{f(b) - f(a)}{g(b) - g(a)}[g(x) - g(a)]$. Clearly, $\varphi(a) = \varphi(b)$. Then by Rolle's theorem, there exists $\xi \in (a, b)$ such that

$$\varphi'(x) = f'(x) - \frac{f(b) - f(a)}{g(b) - g(a)}g'(x) = 0,$$

and the theorem follows. $\square$

Theorem 5 (Taylor's theorem). *Let f be an n -times differentiable function on I whose k -th derivative is denoted by $f^{(k)}$ for $0 \leq k \leq n$. Then there exists $\xi \in (a, x) \subseteq I$ such that*

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n)}(\xi)}{n!} (x-a)^n$$

Proof. Define $p(x)$ and $q(x)$ by $f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (x-a)^k$ and $q(x) = (x-a)^n$. Then $p^{(k)}(a) = 0$ and $q^{(k)}(a) = 0$ for $0 \leq k \leq n-1$. By Mean Value Theorem, there exists a sequence ξ_i in which $\xi_0 = x$ and $\xi_{i+1} \in (a, \xi_i)$ such that

$$\begin{aligned} \frac{p(x)}{q(x)} &= \frac{p(x) - p(a)}{q(x) - q(a)} = \frac{p'(\xi_1)}{q'(\xi_1)} \\ \frac{p'(\xi_1)}{q'(\xi_1)} &= \frac{p'(\xi_1) - p'(a)}{q'(\xi_1) - q'(a)} = \frac{p''(\xi_2)}{q''(\xi_2)} \\ &\vdots \end{aligned}$$

Inductively, we have the equation

$$\frac{p(x)}{q(x)} = \frac{p^{(n)}(\xi)}{q^{(n)}(\xi)} = \frac{f^{(n)}(\xi)}{n!}$$

and the theorem follows. $\square$

Theorem 6 (Chain rule, assumed continuity on differential). *Let $f : I \rightarrow R$ and $g : D \rightarrow I$ be two continuously differentiable functions, whose derivatives are denoted by f' and g' respectively. Let $h := f \circ g : D \rightarrow R$ be a composite function. Then h is differentiable on D and its derivative is defined as*

$$h' = (f' \circ g)g'$$

Weak proof. Consider the Taylor expansions of f and g near $b \in I$ and $a \in D$,

$$\begin{aligned} f(y) &= f(b) + f'(b)(y-b) + \varepsilon_f(y-b) \\ g(x) &= g(a) + g'(a)(x-a) + \varepsilon_g(x-a) \end{aligned}$$

where $\varepsilon_f = f'(\xi_f) - f'(b)$ and $\varepsilon_g = g'(\xi_g) - g'(a)$. Identifying $b = g(a)$, we have

$$\begin{aligned} f(g(x)) &= f(g(a)) + f'(g(a))(g(x) - g(a)) + \varepsilon_f(g(x) - g(a)) \\ &= f(g(a)) + f'(g(a))g'(a)(x-a) + \varepsilon_g(x-a)f'(g(a)) + \varepsilon_f(g(x) - g(a)) \\ \frac{f(g(x)) - f(g(a))}{x-a} &= f'(g(a))g'(a) + \varepsilon_g f'(g(a)) + \varepsilon_f \frac{g(x) - g(a)}{x-a} \\ (f \circ g)'(a) &= \lim_{x \rightarrow a} [g'(\xi_g)f'(g(a)) + f'(\xi_f)g'(a) - f'(g(a))g'(a)] \end{aligned}$$

By the continuity of f and g , we know that

$$\lim_{x \rightarrow a} \xi_g = a \text{ and } \lim_{x \rightarrow a} \xi_f = \lim_{y \rightarrow b} \xi_f = g(a)$$

by Squeeze theorem, which concludes that

$$\lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a} = f'(g(a))g'(a).$$

□

Remark. *The second proof relies on the continuity of the derivatives of both functions. If such continuity is not guaranteed, the proof is invalid.*

Counter-example. Put $f(x) = \int_0^x Q(x)dx$ where Q is the Cantor function.